

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester V Examination, 2022-23

Paper No. MTB-501: Mathematical Analysis

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory.)

(a) Show that the limit of a convergent sequence is unique. [2]

(b) Show that the sequence $(x_n)_{n=1}^{\infty}$, where $x_n = \left(1 + \frac{1}{n}\right)^n$, is monotonically increasing. [2](c) Determine the values of x for which the series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ is absolutely convergent. [2](d) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{\cos(1/n)}{n^2}$. [2](e) Let $f(x)$ be defined by:

$$f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

Prove that f is Riemann integrable on $\mathbb{R}$ (on any closed bounded interval). [2](f) Let $f(x) = x$ on $[0, 1]$. If $P = \left\{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\right\}$ is a partition of $[0, 1]$, then find:(i) $L(P, f)$, (ii) $U(P, f)$, (iii) $\int_0^1 f dx$, (iv) $\int_0^1 \bar{f} dx$. [4](g) If $f(x)$ and $g(x)$ are Riemann integrable on $[a, b]$ such that $f(x) \leq g(x)$ for all $x \in [a, b]$, show that $\int_a^b f dx \leq \int_a^b g dx$. [2](h) Test the convergence of $\int_0^1 \frac{x}{1-x} dx$. [2](i) Test the convergence of $\int_0^{\infty} e^{-x^2} dx$. [2](j) Show that $\int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$. [2]**Question 2.**

(a) Prove that the necessary and sufficient condition for the convergence of a monotonic sequence of real numbers is that it is bounded. [6]

- (b) Prove that if a sequence $(x_n)_{n=1}^{\infty}$ of positive numbers converges to a limit l , then:

$$\lim_{n \rightarrow \infty} (x_1 x_2 \cdots x_n)^{1/n} = l$$

Deduce that $\lim_{n \rightarrow \infty} n^{1/n} = 1$. [6]

Question 3.

- (a) State and prove Cauchy's general principle for the convergence of sequences of real numbers. [6]
- (b) Let $\sum_{n=1}^{\infty} a_n$ be a positive term series and let $\lim_{n \rightarrow \infty} (a_n)^{1/n} = l$. Then the series is:
- (i) convergent if $l < 1$;
 - (ii) divergent if $l > 1$;
 - (iii) the test fails if $l = 1$.

Prove this result (Cauchy's root test). [6]

Question 4.

- (a) Test the convergence of the series $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$. [6]
- (b) Define absolute and conditional convergence of a series. Show that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^p}$ is absolutely convergent for $p > 1$, but only conditionally convergent for $0 < p \leq 1$. [6]
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Question 5.

- (a) Prove that a necessary and sufficient condition for the Riemann integrability of a bounded function f is that for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every partition P of $[a, b]$ with norm $\|P\| < \delta$:

$$U(P, f) - L(P, f) < \varepsilon$$

[6]

- (b) Prove that every monotonic function on $[a, b]$ is Riemann integrable on $[a, b]$. Give an example of a Riemann integrable function on $[a, b]$ which is not monotonic on $[a, b]$. [6]
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Question 6.

- (a) Show that the function defined by:

$$f(x) = \frac{1}{n+1} \quad \text{for } \frac{1}{n+1} \leq x < \frac{1}{n}, \quad n = 1, 2, 3, \dots$$

is Riemann integrable on $[0, 1]$. Also show that $\int_0^1 f(x) dx = \frac{\pi^2}{6} - 1$. [6]

- (b) If $\int_a^b f(x) dx$ and $\int_a^b g(x) dx$ both exist and $g(x)$ keeps the same sign throughout the interval of integration, then there exists a number μ lying between the bounds of $f(x)$ such that:

$$\int_a^b f(x)g(x) dx = \mu \int_a^b g(x) dx$$

(State and prove the Second Mean Value Theorem for integrals.) [6]

Question 7.

- (a) State and prove Dirichlet's test for improper integrals. [6]
- (b) State and prove the Limit Comparison Test (LCT) for improper integrals. Examine the convergence of $\int_0^\infty \frac{\sin^p x}{x} dx$ for $p > 1$. [6]